

Aufgabe 1.1

```
SELECT
    o.order_year,
    c.region,
    SUM(o.net_amount) AS total_net_revenue,
    COUNT(o.order_id) AS total_orders,
    AVG(o.net_amount) AS avg_order_value
FROM orders o
JOIN customers c
    ON o.customer_id = c.customer_id
GROUP BY
    o.order_year,
    c.region
ORDER BY
    o.order_year,
    c.region;
```

Aufgabe 1.2

```
SELECT
    o.customer_id,
    c.region,
    c.acquisition_chan,
    SUM(o.net_amount) AS total_spent,
    COUNT(o.order_id) AS total_orders
FROM orders o
JOIN customers c
    ON o.customer_id = c.customer_id
GROUP BY
    o.customer_id,
    c.region,
    c.acquisition_chan
ORDER BY
    total_spent DESC
LIMIT 10;
```

Aufgabe 1.3

```
SELECT
    oi.category,
    SUM(oi.line_total) AS total_revenue,
    AVG(p.margin_pct) AS avg_margin_pct,
    COUNT(DISTINCT CASE
        WHEN o.is_returned = 1
        THEN o.order_id
    END) * 1.0
    / COUNT(DISTINCT o.order_id) AS return_rate
FROM order_items oi
JOIN products p
    ON oi.product_id = p.product_id
JOIN orders o
    ON oi.order_id = o.order_id
GROUP BY
    oi.category
ORDER BY
    total_revenue DESC;
```

Aufgabe 1.4

```
SELECT
    COUNT(*) AS high_value_customers,
    SUM(total_spent) AS revenue_from_high_value_customers,
    ROUND(
        SUM(total_spent) * 1.0 /
        (SELECT SUM(net_amount) FROM orders),
        4
    ) AS revenue_share
FROM
    (
        SELECT
            customer_id,
            SUM(net_amount) AS total_spent
```

```

FROM orders
GROUP BY customer_id
HAVING SUM(net_amount) >
(
    SELECT AVG(total_spent)
    FROM
    (
        SELECT
            customer_id,
            SUM(net_amount) AS total_spent
        FROM orders
        GROUP BY customer_id
    ) customer_spending
)
) customer_totals;

```

Aufgabe 1.5

```

SELECT
    channel,
    SUM(budget) AS total_budget,
    SUM(attributed_reven) AS total_revenue,
    ROUND(
        SUM(attributed_reven) * 1.0 /
        NULLIF(SUM(budget), 0),
        2
    ) AS roi
FROM marketing
GROUP BY
    channel
ORDER BY
    roi DESC;

```